ADIEL RUM

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EXECUTIVE SUMMARY

Mathematics undergraduate at **Institut Teknologi Bandung (GPA 3.96)**, minoring in **Computer Science**, specializing in **operations research, data analytics, and supply chain modeling**. Proven track record in developing **demand forecasting models, resource optimization frameworks, and interactive KPI dashboards** for corporate and regulatory stakeholders. Proficient in **Advanced MS Excel** (data reconciliation, pivot reporting), **Power BI, Python, and R**, with demonstrated capability in operational tracking, discrepancy detection, and process efficiency.

EDUCATION

INSTITUT TEKNOLOGI BANDUNG Bandung, Jawa Barat

Bachelors in **Mathematics**, with Minors in **Computer Science** 2023 - 2027 (Projected)

Cumulative GPA: **3.96**/4.0 (125/144 Credits, Transcript)
Relevant Coursework: Mathematical Optimization, Data Analysis, Advanced Time Series Analysis, Operations Research, Numerical Methods, Data Mining, Machine Learning
Awards: Dean's List for 6 consecutive semesters, Bank Syariah Indonesia Scholarship Awardee

SKILLS

Operational Analytics & Modeling: Operations Research, Mathematical Optimization, Demand Forecasting, Time Series Analysis, Lead-Time & Resource Optimization, Anomaly & Discrepancy Detection.
Business Intelligence & Reporting: Advanced MS Excel (Pivot Tables, XLOOKUP, Data Validation & Cleansing, Automated Trackers), Power BI, Google Looker Studio, Interactive Dashboards.
Programming & Tools: Python (Pandas, NumPy, SciPy), R, JavaScript, Git.
Languages: English (Professional Working Proficiency), Indonesian (Native).

WORK EXPERIENCE

Data Analyst Intern (Department of Banking Supervision Quality Control and Development) Jan 2026 - Sep 2026

OTORITAS JASA KEUANGAN (INDONESIA FINANCIAL SERVICES AUTHORITY) Jakarta Pusat, DKI Jakarta

- Audited, validated, and reconciled complex multi-source datasets across 105 entities (600+ records), identifying data discrepancies and ensuring 100% data integrity for supervisory reporting.
- Developed predictive and statistical models to forecast performance metrics with 90% accuracy, supporting forward-looking risk and variance assessments.
- Designed and deployed interactive executive dashboards (Power BI) to monitor risk indicators and operational metrics, streamlining reporting workflows.
- Prepared ad-hoc analytical reports and synthesized weekly meeting minutes for senior leadership, translating quantitative findings into operational decisions.

Research Assistant Apr 2026 - Present

CENTRE OF PREDICTIVE RISK SIMULATION AND MODELLING (PRISM ITB) Bandung, Jawa Barat

- Co-authored quantitative research proposals, securing IDR 30M in research funding for predictive modeling and simulation initiatives.
- Executed advanced quantitative analytics and simulation methodologies to solve complex data-driven optimization problems.
- Designed and deployed interactive analytical dashboards to track key model outcomes and communicate results effectively to multi-disciplinary stakeholders.

PROJECTS

Assistant Researcher (In collaboration with PT. XYZ, Authorized Honda Motor Dealer) Jul 2026 - Present

PREDICT THEN OPTIMIZE FRAMEWORK FOR POINT OF SALE DETERMINATION Bandung, Jawa Barat

- Performed demand forecasting for prospective distribution and point of sale (POS) nodes by analyzing historical trends using statistical time series and machine learning.
- Formulated a multi-objective mathematical optimization model to identify optimal facility locations, balancing customer catchment and service coverage against operational costs.
- Implemented multi-criteria decision algorithms (NSGA-II) to deliver trade-off scenarios and actionable location recommendations aligned with business constraints.

Lead Researcher (In collaboration with IDSC Pte Ltd) Jan 2026 - Jul 2026

RESOURCE-CONSTRAINED PROJECT CRASHING OPTIMIZATION Bandung, Jawa Barat

- Formulated an operational time–cost tradeoff optimization model utilizing production functions to minimize project completion lead time under labor and overtime constraints.
- Evaluated bottleneck activities and operational sensitivity, delivering optimal resource allocation schedules to mitigate schedule slippage and cost overrun risks.
- Presented findings at the Industrial Mathematics Week 2026 Conference and submitted full manuscript for journal publication.

ORGANIZATIONAL EXPERIENCE

Head of Data Science Competition (Mathematics Challenge Festival 2026) Jun 2025 – May 2026

HIMPUNAN MAHASISWA MATEMATIKA ITB (HIMATIKA ITB) Bandung, Jawa Barat

- Directed a 10-member operational committee to organize a national data science competition, driving outreach strategies that attracted over 950 participants nationwide.
- Coordinated with industry professionals and academic advisors to design end-to-end competition workflows, evaluation rubrics, and data tracking pipelines.